1. Random variables

Definition 1.1 (Expectation)**.** The expectation of a random variable X on a probability space $(\Omega, \mathcal{E}, \mathbb{P})$ is

$$\mathbb{E}[X] = \int X \, \mathrm{d}\mathbb{P},$$

whenever well-defined.

Remark. We require at least one of $\mathbb{E}[X^+], \mathbb{E}[X^-]$ to be finite.

Proposition 1.2. *For any* $A \in \mathcal{E}$,

$$\mathbb{P}(X \in A) = \mathbb{E}[\mathbf{1}_A(X)].$$

Theorem 1.3 (Markov)**.** *Let* $X \geq 0$. *For all* $a > 0$,

$$\mathbb{P}(X > a) \leq \frac{\mathbb{E}[X]}{a}.$$

Corollary 1.3.1 (Chebyshev)**.** *Let* $\mathbb{E}[X] = \mu$, $\operatorname{var}[X] = \sigma^2$. *Then,*

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

A. Appendix

Proof of Theorem 1.3.

$$\begin{aligned}
\mathbb{P}(X > a) &= \mathbb{E}\left[\mathbf{1}_{(a,\infty)}(X)\right] && \text{(Proposition 1.2)} \\
&\leq \mathbb{E}\left[\left(\frac{X}{a}\right)\mathbf{1}_{(a,\infty)}(X)\right] \\
&\leq \frac{\mathbb{E}[X]}{a}.
\end{aligned}$$

$\square$